Zero-inflated models: Poisson, Binomial, negative Binomial and BetaBinomial

Parametrisation

There is support two types of zero-inflated models, which we name type 0 and type 1. These are defined for both the Binomial, the Poisson, the censored Poisson, the negative Binomial and BetaBinomial likelihood. For simplicity we will describe only the Poisson as the other cases are similar.

Type 0

The (type 0) likelihood is defined as

$$\text{Prob}(y \mid \ldots) = p \times 1_{[y=0]} + (1-p) \times \text{Poisson}(y \mid y > 0)$$

where p is a hyperparameter where

$$p = \frac{\exp(\theta)}{1+\exp(\theta)}$$

and θ is the internal representation of p; meaning that the initial value and prior is given for θ. This is model is called `zeroinflatedpoisson0` (and `zeroinflatedbinomial0`).

Type 1

The (type 1) likelihood is defined as

$$\text{Prob}(y \mid \ldots) = p \times 1_{[y=0]} + (1-p) \times \text{Poisson}(y)$$

where p is a hyperparameter where

$$p = \frac{\exp(\theta)}{1+\exp(\theta)}$$

and θ is the internal representation of p; meaning that the initial value and prior is given for θ. This is model is called `zeroinflatedpoisson1` (and `zeroinflatedbinomial1`).

Link-function

As for the Poisson, the Binomial the negative Binomial and the BetaBinomial.

Hyperparameters

For Poisson and the Binomial, there is one hyperparameter; where

$$p = \frac{\exp(\theta)}{1+\exp(\theta)}$$

and the prior and initial value is is given for θ.

For the negative Binomial and BetaBinomial, there are two hyperparameters. The overdispersion parameter n for the negative Binomial is represented as

$$\theta_1 = \log(n)$$

and the prior is defined on θ_1. The zero-inflation parameter p, is represented as

$$p = \frac{\exp(\theta_2)}{1+\exp(\theta_2)}$$

and the prior and initial value is is given for θ_2. For the BetaBinomial it is similar.

Specification

- `family="zeroinflatedbinomial0"`
- `family="zeroinflatedbinomial1"`
- `family="zeroinflatednbinomial0"`
- `family="zeroinflatednbinomial1"`
- `family="zeroinflatedpoisson0"`
- `family="zeroinflatedpoisson1"`
- `family="zeroinflatedcenpoisson0"`
- `family="zeroinflatedcenpoisson1"`
- `family="zeroinflatedbetabinomial0"`
- `family="zeroinflatedbetabinomial1"`
- Required arguments: As for the Binomial, the negative Binomial, BetaBinomial and Poisson likelihood.

Hyperparameter spesification and default values

Zeroinflated Binomial Type 0

doc `Zero-inflated Binomial, type 0`

hyper

theta

hyperid `90001`
name `logit probability`
short.name `prob`
output.name `zero-probability parameter for zero-inflated binomial_0`
output.name.intern `intern zero-probability parameter for zero-inflated binomial_0`
initial `-1`
fixed `FALSE`
prior `gaussian`
param `-1 0.2`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) exp(x) / (1 + exp(x))`

survival `FALSE`

discrete `FALSE`

link `default logit loga cauchit probit cloglog ccloglog loglog robit sn`

pdf `zeroinflated`

Zeroinflated Binomial Type 1

doc Zero-inflated Binomial, type 1

hyper

theta

hyperid 91001
name logit probability
short.name prob
output.name zero-probability parameter for zero-inflated binomial_1
output.name.intern intern zero-probability parameter for zero-inflated binomial_1
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete FALSE

link default logit loga cauchit probit cloglog ccloglog loglog robit sn

pdf zeroinflated

Zeroinflated NegBinomial Type 0

doc Zero inflated negBinomial, type 0

hyper

theta1

hyperid 95001
name log size
short.name size
output.name size for nbinomial_0 zero-inflated observations
output.name.intern log size for nbinomial_0 zero-inflated observations
initial 2.30258509299405
fixed FALSE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta2

hyperid 95002
name logit probability
short.name prob
output.name zero-probability parameter for zero-inflated nbinomial_0

output.name.intern intern zero-probability parameter for zero-inflated nbinomial_0
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete FALSE

link default log

pdf zeroinflated

Zeroinflated NegBinomial Type 1

doc Zero inflated negBinomial, type 1

hyper

theta1
hyperid 96001
name log size
short.name size
output.name size for nbinomial_1 zero-inflated observations
output.name.intern log size for nbinomial_1 zero-inflated observations
initial 2.30258509299405
fixed FALSE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta2
hyperid 96002
name logit probability
short.name prob
output.name zero-probability parameter for zero-inflated nbinomial_1
output.name.intern intern zero-probability parameter for zero-inflated nbinomial_1
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete FALSE

link default log

pdf zeroinflated

Zeroinflated BetaBinomial Type 0

doc Zero-inflated Beta-Binomial, type 0

hyper

theta1

hyperid 88001
name overdispersion
short.name rho
output.name rho for zero-inflated betabinomial_0
output.name.intern rho_intern for zero-inflated betabinomial_0
initial 0
fixed FALSE
prior gaussian
param 0 0.4
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta2

hyperid 88002
name logit probability
short.name prob
output.name zero-probability parameter for zero-inflated betabinomial_0
output.name.intern intern zero-probability parameter for zero-inflated betabinomial_
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete TRUE

link default logit loga cauchit probit cloglog ccloglog loglog robit sn

pdf zeroinflated

Zeroinflated BetaBinomial Type 1

doc Zero-inflated Beta-Binomial, type 1

hyper

theta1

hyperid 89001
name overdispersion
short.name rho
output.name rho for zero-inflated betabinomial_1

output.name.intern rho_intern for zero-inflated betabinomial_1
initial 0
fixed FALSE
prior gaussian
param 0 0.4
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta2
hyperid 89002
name logit probability
short.name prob
output.name zero-probability parameter for zero-inflated betabinomial_1
output.name.intern intern zero-probability parameter for zero-inflated betabinomial_
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete TRUE

link default logit loga cauchit probit cloglog ccloglog loglog robit sn

pdf zeroinflated

Zeroinflated Poisson Type 0

doc Zero-inflated Poisson, type 0

hyper

theta
hyperid 85001
name logit probability
short.name prob
output.name zero-probability parameter for zero-inflated poisson_0
output.name.intern intern zero-probability parameter for zero-inflated poisson_0
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete FALSE

link default log

pdf zeroinflated

Zeroinflated Poisson Type 1

doc `Zero-inflated Poisson, type 1`

hyper

theta

hyperid `86001`
name `logit probability`
short.name `prob`
output.name `zero-probability parameter for zero-inflated poisson_1`
output.name.intern `intern zero-probability parameter for zero-inflated poisson_1`
initial `-1`
fixed `FALSE`
prior `gaussian`
param `-1 0.2`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) exp(x) / (1 + exp(x))`

survival `FALSE`

discrete `FALSE`

link `default log`

pdf `zeroinflated`

Zeroinflated Censored Poisson Type 0

doc `Zero-inflated censored Poisson, type 0`

hyper

theta

hyperid `87101`
name `logit probability`
short.name `prob`
output.name `zero-probability parameter for zero-inflated poisson_0`
output.name.intern `intern zero-probability parameter for zero-inflated poisson_0`
initial `-1`
fixed `FALSE`
prior `gaussian`
param `-1 0.2`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) exp(x) / (1 + exp(x))`

survival `FALSE`

discrete `FALSE`

link `default log`

pdf `zeroinflated`

Zeroinflated Censored Poisson Type 1

doc `Zero-inflated censored Poisson, type 1`

hyper

theta

hyperid `87201`
name `logit probability`
short.name `prob`
output.name `zero-probability parameter for zero-inflated poisson_1`
output.name.intern `intern zero-probability parameter for zero-inflated poisson_1`
initial `-1`
fixed `FALSE`
prior `gaussian`
param `-1 0.2`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) exp(x) / (1 + exp(x))`

survival `FALSE`

discrete `FALSE`

link `default log`

pdf `zeroinflated`

Example

In the following example we estimate the parameters in a simulated example for both type 0 and type 1.

Poisson

```
## type 0
n=100
a = 1
b = 1
z = rnorm(n)
eta = a + b*z
p = 0.2
E = sample(c(1,5,10,15), size=n, replace=TRUE)
lambda = E*exp(eta)

## first sample y|y>0
y = rpois(n, lambda = lambda)
is.zero = (y == 0)
while(sum(is.zero) > 0)
{
    y[is.zero] = rpois(sum(is.zero), lambda[is.zero])
    is.zero = (y == 0)
}
```

```
## then set some of these to zero
y[ rbinom(n, size=1, prob=p) == 1 ] = 0

data = list(y=y,z=z)
formula = y ~ 1+z
result0 = inla(formula, family = "zeroinflatedpoisson0", data = data, E=E)
summary(result0)

## type 1
y = rpois(n, lambda = lambda)
y[ rbinom(n, size=1, prob=p) == 1 ] = 0
data = list(y=y,z=z)
formula = y ~ 1+z
result1 = inla(formula, family = "zeroinflatedpoisson1", data = data, E=E)
summary(result1)
```

Binomial

```
## type 0
n=100
a = 1
b = 1
z = rnorm(n)
eta = a + b*z
p = 0.2
Ntrials = sample(c(1,5,10,15), size=n, replace=TRUE)
prob = exp(eta)/(1 + exp(eta))

y = rbinom(n, size = Ntrials, prob = prob)
is.zero = (y == 0)
while(sum(is.zero) > 0)
{
    y[is.zero] = rbinom(sum(is.zero), size = Ntrials[is.zero], prob = prob[is.zero])
    is.zero = (y == 0)
}
y[ rbinom(n, size=1, prob=p) == 1 ] = 0
data = list(y=y,z=z)
formula = y ~ 1+z
result0 = inla(formula, family = "zeroinflatedbinomial0", data = data, Ntrials = Ntrials)
summary(result0)

## type 1
y = rbinom(n, size = Ntrials, prob = prob)
y[ rbinom(n, size=1, prob=p) == 1 ] = 0
data = list(y=y,z=z)
formula = y ~ 1+z
result1 = inla(formula, family = "zeroinflatedbinomial1", data = data, Ntrials=Ntrials)
summary(result1)
```

Advanced example

In the following example we estimate the parameters in a simulated example for a type0 likelihood, where one linear predictor enters the zero-probability and one other linear predictor enters the non-zero Poisson for example. The same trick can be used for other models of type0. The trick is that the likelihood

$$p^* 1_{[y=0]} + (1 - p^*) P(y|y > 0)$$

can be reformulated as a Bernoulli likelihood for the "class"-variable

$$z = \begin{cases} 1, & \text{if } y = 0 \\ 0, & \text{if } y > 0. \end{cases}$$

where p^* is the probability for success, and zero-inflated type0 likelihood (with fixed $p = 0$) for those $y > 0$. Since p^* and the linear predictor in P is separated into two likelihoods, we can apply one linear predictor to each one, hence extend the basic model to cases where p^* also depends on a linear predictor. Here is a small simulated example doing this.

```
require(INLA)

n = 100
a = 0.5
b = 1.5
x1 = rnorm(n, sd = 0.5)

eta.z = -a - b*x1
z = rbinom(n, 1, inla.link.logit(eta.z, inverse=TRUE))
n.y = sum(z)

x2 = rnorm(n.y, sd = 0.5)
eta.y = a + b*x2
lambda = exp(eta.y)
y = rpois(n.y, lambda)

is.zero = (y == 0)
while(sum(is.zero) > 0)
{
   y[is.zero] = rpois(sum(is.zero), lambda[is.zero])
   is.zero = (y == 0)
}

Y = matrix(NA, n + n.y, 2)
Y[1:n, 1] = z
Y[n + 1:n.y, 2] = y

form = Y ~ 0 + mu.z + mu.y + cov.z + cov.y
ldat = list(
        Y=Y,
        mu.z=rep(1:0, c(n, n.y)),
        mu.y=rep(0:1, c(n, n.y)),
        cov.z=c(x1, rep(NA,n.y)),
        cov.y=c(rep(NA, n), x2))
```

```
res <- inla(form, data=ldat,
            family=c('binomial', 'zeroinflatedpoisson0'),
            control.family=list(
                    list(),
                    list(hyper = list(
                                prob = list(
                                        initial = -20,
                                        fixed = TRUE)))))
round(res$summary.fix, 4)
```

Notes

None.

Extentions

There are some extentions available which currently is only implemented for the cases where its needed/requested.

Type 2 Is like Type 1 but where (for the Poisson)

$$p = 1 - \left(\frac{E\exp(x)}{1 + E\exp(x)}\right)^{\alpha}$$

where $\alpha > 0$ is the hyperparameter instead of p (and $E\exp(x)$ is the mean). Available for Poisson as `zeroinflatedpoisson2`, for binomial as `zeroinflatedbinomial2` and for the negative binomial as `zeroinflatednbinomial2`.

The internal representation is $\theta = \log(\alpha)$ and prior is defined on $\log(\alpha)$.

Zeroinflated Poisson Type 2

doc `Zero-inflated Poisson, type 2`

hyper

theta

hyperid `87001`
name `log alpha`
short.name `a`
output.name `zero-probability parameter for zero-inflated poisson_2`
output.name.intern `intern zero-probability parameter for zero-inflated poisson_2`
initial `0.693147180559945`
fixed `FALSE`
prior `gaussian`
param `0.693147180559945 1`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

survival `FALSE`

discrete `FALSE`

link `default log`

pdf `zeroinflated`

Zeroinflated Binomial Type 2

doc Zero-inflated Binomial, type 2

hyper

theta

hyperid 92001
name alpha
short.name alpha
output.name zero-probability parameter for zero-inflated binomial_2
output.name.intern intern zero-probability parameter for zero-inflated binomial_2
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x)
from.theta function(x) exp(x)

survival FALSE

discrete FALSE

link default logit loga cauchit probit cloglog ccloglog loglog robit sn

pdf zeroinflated

Zeroinflated Negative Binomial Type 2

doc Zero inflated negBinomial, type 2

hyper

theta1

hyperid 99001
name log size
short.name size
output.name size for nbinomial zero-inflated observations
output.name.inter log size for nbinomial zero-inflated observations
initial 2.30258509299405
fixed FALSE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta2

hyperid 99002
name log alpha
short.name a
output.name parameter alpha for zero-inflated nbinomial2

output.name.intern parameter alpha.intern for zero-inflated nbinomial2
initial 0.693147180559945
fixed FALSE
prior gaussian
param 2 1
to.theta function(x) log(x)
from.theta function(x) exp(x)

survival FALSE

discrete FALSE

link default log

pdf zeroinflated

Zeroinflated Negative Binomial Type 1 Strata 2

doc Zero inflated negBinomial, type 1, strata 2

hyper

theta1

hyperid 97001
name log size
short.name size
output.name size for zero-inflated nbinomial_1_strata2
output.name.intern log size for zero-inflated nbinomial_1_strata2
initial 2.30258509299405
fixed FALSE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta2

hyperid 97002
name logit probability 1
short.name prob1
output.name zero-probability1 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability1 for zero-inflated nbinomial_1_strata2
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta3

hyperid 97003

name logit probability 2
short.name prob2
output.name zero-probability2 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability2 for zero-inflated nbinomial_1_strata2
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta4

hyperid 97004
name logit probability 3
short.name prob3
output.name zero-probability3 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability3 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta5

hyperid 97005
name logit probability 4
short.name prob4
output.name zero-probability4 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability4 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta6

hyperid 97006
name logit probability 5
short.name prob5
output.name zero-probability5 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability5 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2

to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta7

hyperid 97007
name logit probability 6
short.name prob6
output.name zero-probability6 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability6 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta8

hyperid 97008
name logit probability 7
short.name prob7
output.name zero-probability7 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability7 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta9

hyperid 97009
name logit probability 8
short.name prob8
output.name zero-probability8 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability8 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta10

hyperid 97010
name logit probability 9
short.name prob9
output.name zero-probability9 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability9 for zero-inflated nbinomial_1_strata2

initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta11

hyperid 97011
name logit probability 10
short.name prob10
output.name zero-probability10 for zero-inflated nbinomial_1_strata2
output.name.intern intern zero-probability10 for zero-inflated nbinomial_1_strata2
initial -1
fixed TRUE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

survival FALSE

discrete FALSE

link default log

pdf zeroinflated

Zeroinflated Negative Binomial Type 1 Strata 3

doc Zero inflated negBinomial, type 1, strata 3

hyper

theta1

hyperid 98001
name logit probability
short.name prob
output.name zero-probability for zero-inflated nbinomial_1_strata3
output.name.intern intern zero-probability for zero-inflated nbinomial_1_strata3
initial -1
fixed FALSE
prior gaussian
param -1 0.2
to.theta function(x) log(x / (1 - x))
from.theta function(x) exp(x) / (1 + exp(x))

theta2

hyperid 98002
name log size 1

short.name size1
output.name size1 for zero-inflated nbinomial_1_strata3
output.name.intern log_size1 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed FALSE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta3

hyperid 98003
name log size 2
short.name size2
output.name size2 for zero-inflated nbinomial_1_strata3
output.name.intern log_size2 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed FALSE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta4

hyperid 98004
name log size 3
short.name size3
output.name size3 for zero-inflated nbinomial_1_strata3
output.name.intern log_size3 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta5

hyperid 98005
name log size 4
short.name size4
output.name size4 for zero-inflated nbinomial_1_strata3
output.name.intern log_size4 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)

from.theta function(x) exp(x)

theta6

hyperid 98006
name log size 5
short.name size5
output.name size5 for zero-inflated nbinomial_1_strata3
output.name.intern log_size5 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta7

hyperid 98007
name log size 6
short.name size6
output.name size6 for zero-inflated nbinomial_1_strata3
output.name.intern log_size6 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta8

hyperid 98008
name log size 7
short.name size7
output.name size7 for zero-inflated nbinomial_1_strata3
output.name.intern log_size7 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta9

hyperid 98009
name log size 8
short.name size8
output.name size8 for zero-inflated nbinomial_1_strata3
output.name.intern log_size8 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405

fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta10
hyperid 98010
name log size 9
short.name size9
output.name size9 for zero-inflated nbinomial_1_strata3
output.name.intern log_size9 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta11
hyperid 98011
name log size 10
short.name size10
output.name size10 for zero-inflated nbinomial_1_strata3
output.name.intern log_size10 for zero-inflated nbinomial_1_strata3
initial 2.30258509299405
fixed TRUE
prior pc.mgamma
param 7
to.theta function(x) log(x)
from.theta function(x) exp(x)

survival FALSE

discrete FALSE

link default log

pdf zeroinflated

0.0.1 Zero and N-inflated Binomial likelihood: type 3

This is the case where

$$\begin{aligned}\text{Prob}(y|\ldots) &= p_0 \times 1_{[y=0]} + \\ &\quad p_N \times 1_{[y=N]} + \\ &\quad (1-p_0-p_N) \times \text{binomial}(y,N,p)\end{aligned}$$

where:

$$p = \frac{\exp(\eta)}{1+\exp(\eta)} \qquad p_0 = \frac{p^{\alpha_0}}{1+p^{\alpha_0}+(1-p)^{\alpha_N}} \qquad p_N = \frac{(1-p)^{\alpha_N}}{1+p^{\alpha_0}+(1-p)^{\alpha_N}}$$

There are 2 hyperparameters, α_0 and α_N, governing zero-inflation where: The zero-inflation parameters α_0 and α_N are represented as $\theta_0 = \log(\alpha_0)$; $\theta_N = \log(\alpha_N)$ and the prior and initial value is given for θ_0 and θ_N respectively.

Here is an example

```
nsim<-10000
x<-rnorm(nsim)
alpha0<-1.5
alphaN<-2.0
p = exp(x)/(1+exp(x))
p0 = p^alpha0 / (1 + p^alpha0 + (1-p)^alphaN)
pN = (1-p)^alphaN / (1 + p^alpha0 + (1-p)^alphaN)
P<-cbind(p0, pN, (1-p0 -pN))
N<-rpois(nsim,20)
y<-rep(0,nsim)
for(i in 1:nsim)
    y[i]<-sum(rmultinom(1,size = 1,P[i,])*c(0,N[i],rbinom(1,N[i],p[i])))
formula = y ~1 + x
r = inla(formula, family = "zeroninflatedbinomial3",  Ntrials = N, verbose = TRUE,
           data = data.frame(y, x))
```

and the default settings

doc `Zero and N inflated binomial, type 3`

hyper

- **theta1**
  - **hyperid** `93101`
  - **name** `alpha0`
  - **short.name** `alpha0`
  - **output.name** `alpha0 parameter for zero-n-inflated binomial_3`
  - **output.name.intern** `intern alpha0 parameter for zero-n-inflated binomial_3`
  - **initial** `1`
  - **fixed** `FALSE`
  - **prior** `loggamma`
  - **param** `1 1`
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`
- **theta2**
  - **hyperid** `93102`
  - **name** `alphaN`
  - **short.name** `alphaN`
  - **output.name.intern** `intern alphaN parameter for zero-n-inflated binomial_3`
  - **output.name** `alphaN parameter for zero-n-inflated binomial_3`
  - **initial** `1`
  - **fixed** `FALSE`
  - **prior** `loggamma`
  - **param** `1 1`

to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

survival `FALSE`

discrete `FALSE`

link `default logit loga cauchit probit cloglog ccloglog loglog robit sn`

pdf `zeroinflated`